

MLOPS IN PRODUCTION · ISSUE 05 OF 08

Reproducible pipelines with MLflow and Kubeflow.

If you cannot rerun last month's model, you cannot debug it.

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Swipe →

What to version

- ☒ Code commit
- ☒ Data snapshot or version
- ☒ Feature definitions
- ☒ Parameters and random seed
- ☒ Container image digest
- ☒ Dependency lockfile
- ☒ Evaluation dataset version



Pipeline steps

1 Ingest

2 Validate

3 Build features

4 Train

5 Evaluate

6 Register

Each step is a container with declared inputs and outputs.

Who does what

MLFLOW

- Experiment tracking
- Model registry and aliases
- Artifact storage
- Library + server

KUBEFLOW PIPELINES

- Orchestration on Kubernetes
- DAG of containers
- Retries and caching
- Runs on your cluster

Log what you need to rerun

PYTHON

```
with mlflow.start_run() as run:
    mlflow.set_tag("git_commit", GIT_SHA)
    mlflow.set_tag("image", IMAGE_DIGEST)
    mlflow.log_params({"seed": 42, **params})
    mlflow.log_input(
        mlflow.data.from_pandas(df, name="train-2026-10-01")
    )
    model = train(df, params)
    mlflow.log_metric("auc", evaluate(model))
```

Reproducibility habits

DO

- ✓ Pin dependencies with a lockfile
- ✓ Snapshot data, never read “latest”
- ✓ Run training in the same image as CI

AVOID

- ✗ pip install in a notebook
- ✗ Unseeded randomness
- ✗ Data paths without versions

KEY TAKEAWAY

Version code, data, environment and evaluation together.

- Seven things to version
- MLflow tracks, Kubeflow orchestrates
- Same image in CI and training

YOUR TURN

Could you rerun the model that is in production right now?

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Next in the series: #06 Feature quality and lineage

Save · Repost

